A Novel Multi-Feature Fusion Network With Spatial Partitioning Strategy and Cross-Attention for Armband-Based Gesture Recognition

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***Abstract*— Effectively integrating the time-space-frequency information of multi-modal signals from armband sensor, including surface electromyogram (sEMG) and accelerometer data, is critical for accurate gesture recognition. Existing approaches often neglect the abundant spatial relationships inherent in multi-channel sEMG signals obtained via armband sensors and face challenges in harnessing the correlations across multiple feature domains. To address this issue, we propose a novel multi-feature fusion network with spatial partitioning strategy and cross-attention (MFN-SPSCA) to improve the accuracy and robustness of gesture recognition. Specifically, a spatiotemporal graph convolution module with a spatial partitioning strategy is designed to capture potential spatial feature of multi-channel sEMG signals. Additionally, we design a cross-attention fusion module to learn and prioritize the importance and correlation of multi-feature domain. Extensive experiment demonstrate that the MFN-SPSCA method outperforms other state-of-the-art methods on self-collected dataset and the Ninapro DB5 dataset. Our work addresses the challenge of recognizing gestures from the multi-modal data collected by armband sensor, emphasizing the importance of integrating time-space-frequency information. Codes are available at https://github.com/ZJUTofBrainIntelligence/MFN-SPSCA.**

***Index Terms*— Gesture recognition, multi-feature fusion, deep learning, spatial partitioning strategy, cross-attention.**

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I. Introduction

GESTURE recognition plays a crucial role in modern neural engineering [1], providing essential technical support for the development of rehabilitation devices such as prosthetics and exoskeletons [2], [3]. At present, gesture recognition sensor forms can be categorized into two types: Non-wearable sensors (e.g., visual sensors) and wearable sensors (e.g., surface electromyography (sEMG) or inertial sensors) [4], [5]. Gesture recognition technology based on visual sensors is limited by experimental environments and is susceptible to factors such as light intensity and occlusion, resulting in poor robustness. On the contrary, wearable sensors-based gesture recognition technology experiences fewer external environmental interferences and can effectively overcome the aforementioned limitations [6], [7], [8]. Among various sensor technologies, the Myo armband [9], which integrates sEMG and inertial measurement sensors, has emerged as a critical tool in neural engineering applications.

The Myo armband's ability to capture detailed physiological signals from multiple muscle groups makes it especially valuable for controlling prosthetic limbs and exoskeletons. Therefore, significant progress has been made in academic research utilizing sEMG signals from Myo armband sensor for gesture recognition. However, practical applications of the related methods still encounter issues such as sensor misplacement, muscle fatigue, and skin perspiration. Obviously, these issues may lead to a decrease in the quality of sEMG signals and result in destructive deterioration in model recognition performance [6], [10]. Numerous studies [11], [12], [13] indicate that sEMG and acceleration (ACC) signals can provide complementary physiological and kinematic information, enhancing the accuracy and stability of gesture recognition. Therefore, a fusion decoding framework for sEMG and ACC signals emerges as a promising solution for gesture recognition. However, the design of the fusion framework encounters two persistent and unresolved challenges. The first challenge lies in achieving multi-feature (multi-modal and multi-domain) semantic decoding of sEMG and ACC signals to obtain a comprehensive feature representation. The second challenge is how to effectively integrate multi-feature semantic information and

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dynamically learn the latent correlations among multi-feature domains. It should be noted that obtaining comprehensive feature representation is an important prerequisite for achieving effective multi-feature fusion.

Since sEMG and ACC signals encompass abundant temporal, frequency, and spatial information. The development of efficacious feature learning framework constitutes a pivotal strategy in addressing the first challenge. Existing feature learning methods are primarily categorized into two types: Machine learning and deep learning. Machine learning methods typically involve three processes: Preprocessing, feature extraction and classification. Commonly used classifiers include k-nearest neighbor (KNN) [13], random forest (RF) [14], linear discriminant analysis (LDA) [15], and others. Kuzborskij et al. [16] extracted seven discriminative time-frequency domain features and tested them using four classifiers on the NinaPro dataset. They found that combining of mean absolute value features with a non-linear SVM model resulted the highest recognition rate for the 52 gesture movements, achieving an accuracy of approximately 80%. Gijsberts et al. [17] separately extracted marginal discrete wavelet transform (mDWT) features from sEMG signals and mean value features from inertial signals. They achieved satisfactory gesture classification performance using a nonlinear kernel regularized least squares classifier. However, traditional machine learning methods involve complex manual feature extraction processes that heavily rely on expert experience, leading to poor generalization performance of the model. Due to the end-to-end learning capabilities, practicality, and adaptability, deep learning methods, have demonstrated superior gesture recognition performance compared to traditional machine learning methods [18], [19], [20], [21]. Shen et al. [22] utilized normalized sEMG signals as inputs for a convolutional neural network (CNN) model, achieving effective differentiation of 52 gestures on the Ninapro DB5 dataset with an accuracy of 74.51%. Koch et al. [23] proposed a non-uniformly stacked recurrent neural network consisting of long short-term memory (LSTM) and convolutional LSTM, achieving a gesture recognition accuracy of 77.8% for amputees and 82.7% for healthy individuals. Nevertheless, single-model methods based on CNN or RNN face challenges in simultaneously learning both short-term and long-term dependencies. Therefore, the study [24] introduced a hybrid model named recurrent convolutional neural network (RCNN), which combines CNN and RNN to analyze sEMG signals. The RCNN model achieves a recognition accuracy of 87.3% for 50 gestures in Ninapro DB2 dataset. Despite enhancing gesture recognition accuracy to some extent, hybrid models integrating CNN and RNN encounter difficulties in capturing spatial information across multi-channel signals. Gesture execution is intricately connected to the synchronized activation and coordination of numerous muscle groups [25]. Accurately decoding gesture movements relies on comprehending the nuanced interactions among these muscle groups. Graph convolutional networks (GCN) possess robust capabilities in modeling spatial information, prompting several studies [26], [27], [28] to utilize them for investigating potential correlated information among multi-channel physiological signals. Nonetheless, current methods based on GCN only concentrate on associative information among nodes, making it difficult to capture high-level spatial semantic information at the regional level. Overall, despite the growing application of deep learning networks in gesture recognition, existing methods struggle to explore comprehensive feature representations, limiting the performance of gesture recognition models.

Developing an effective fusion strategy is essential to addressing the second challenge. Current fusion methods primarily consist of feature-level fusion and decision-level fusion [29]. Li et al. [30] introduced a component-based method for sign language recognition, employing sEMG and ACC features to illustrate the hand shape, direction, and movement. They then employed a Gaussian mixture model for effective decision-level fusion in classification. Wei et al. [31] employed the classical multi-view learning approach (MV-CNN) to achieve hand gesture recognition (HGR) using multi-modal signals. MV-CNN considers various feature sets extracted from sEMG and inertial measurement unit signals to be different views of gesture movements. The approach employs multiple-stream CNNs to model these views parallelly, and finally adopts a decision-level fusion approach to integrate the outputs of multiple CNNs, resulting in the final classification result. Although the aforementioned studies have produced favorable results, decision-level fusion is not a strictly end-to-end process, which relies on the performance of independent models and lacks interpretability. In contrast, feature-level fusion prioritizes the integration of key features to exploit the complementary interactions among multi-feature information fully. Feature-level fusion methods can be further classified into concatenation fusion [18], additive fusion [32], and self-attention fusion [42]. Krasoulis et al. [15] extracted seven sEMG features and nine IMU features, concatenating them along the feature dimension before inputting them into LDA for classification. Liu et al. [33] extracted time-domain features from sEMG and ACC signals, and further employed a feature accumulation approach for fusion, enabling continuous recognition of 17 multifunctional movements involving fingers and wrists. Duan et al. [34] concatenated feature maps of sEMG and ACC signals along the channel dimension and then fed them into a self-attention mechanism for fusion classification. They attained a recognition accuracy of 87.04% for 41 gestures on the Ninapro DB5 dataset. In conclusion, although many traditional feature-level fusion methods are widely applied to multi-modal fusion tasks, they have failed to sufficiently address the potential dependencies and complex interactions among feature domains. As a consequence, these limitations have imposed constraints on the recognition efficacy of current models.

In conclusion, although current methods have demonstrated competitiveness, they still struggle to accurately model gesture recognition tasks based on multi-modal data from armbands for the following reasons: *1. Complexity of the Data:* sEMG and ACC data, which are commonly used for gesture recognition, are highly complex and non-stationary. Furthermore, sEMG and ACC signals are multi-dimensional, representing different aspects of gesture movement (temporal, spatial, and frequency-based features). Capturing meaningful semantic

information from such diverse features requires sophisticated techniques that can handle the inherent variability and complex interdependencies of the data. *2. Limitations of Existing Algorithms:* Many traditional algorithms for gesture recognition struggle to effectively combine these heterogeneous features. Specifically, existing methods have not fully explored the inherent spatial information between sEMG channels collected from armband sensors. In addition, conventional models either focus on a single modality or fail to capture the interdependencies between features across different domains. Therefore, the performance of existing algorithms is often limited in real-world applications.

To address the aforementioned issues, this paper proposes a novel deep learning-based multibranch network named MFN-SPSCA to simultaneously pursue the accuracy and quite robustness of armband-based gesture recognition. MFN-SPSCA consists of several modules: Recurrent multiscale convolution module (RMSCM), spatiotemporal graph convolution module with spatial partitioning strategy (STGCM-SPS), continuous wavelet transform-convolution module (CWT-CM), cross-attention fusion module (CAFM), and classification module (CM). MFN-SPSCA aims to extract discriminative temporal, spatial, and frequency-domain features from multi-modal data using RMSCM, STGCM-SPS, and CWT-CM, respectively. Additionally, CAFM is utilized to learn and integrate potential correlation information among multiple feature sets. Subsequently, the fused features are fed into the CM to yield precise prediction results. Moreover, the efficacy of the proposed model is validated through performance comparisons with state-of-the-art methods and ablation studies. The primary contributions of this paper are summarized as follows:

1) We propose a spatial partitioning strategy integrated into the GCN, which effectively captures the inherent spatial information between multiple sEMG channels. In addition, this approach enables the model to capture the functional connectivity between muscles, providing a more detailed representation of neuromuscular control mechanisms.
2) We propose a feature fusion module based on a cross-attention fusion method, which effectively integrates multi-feature information across modalities and time-space-frequency domains, leading to a significant enhancement in the model's classification performance.
3) MFN-SPSCA exhibits outstanding gesture recognition and multi-feature information integration capabilities. Experimental results demonstrate the superiority of the MFN-SPSCA method compared with state-of-the-art methods across both our self-collected dataset and the publicly Ninapro DB5 dataset.

The remaining sections of this paper are organized as follows. Section II introduces the datasets used in this study and the preprocessing steps. Section III presents the proposed method and its details. Section IV showcases the experimental results. Section V draws comprehensive discussions. Section VI draws conclusions and prospects for future work.

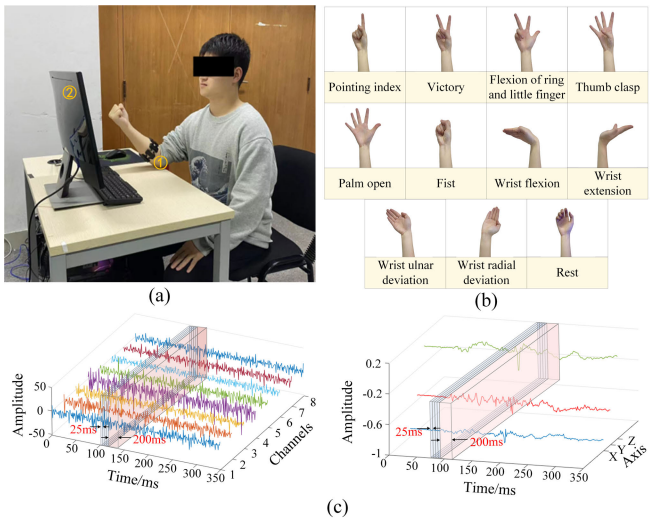


Fig. 1. Experimental setup. (a) Experimental scene. ①: Myo armband, ②: Terminal device. (b) 11 gesture movements. (c) The collected sEMG and ACC signals.

II. Experimental Setup and Materials

A. Datasets

Considering that the proposed model is specifically designed for multi-modal data from armband sensor, this paper evaluates the performance of the proposed model using the publicly Ninapro DB5 dataset. Additionally, a self-collected dataset, named MyoData, is established based on a single armband sensor. The details of the two datasets are described below.

MyoData: A total of 17 healthy subjects (11 males, age: 24 ± 2.3 years, weight: 65 ± 8.7 kg, height: 175 ± 6.5 cm. 6 females, age: 24 ± 1.6 years, weight: 48 ± 9.7 kg, height: 165 ± 5.3 cm) were recruited. Approval of all ethical and experimental procedures and protocols wasgranted by Ethics Committee of Zhejiang University of Technology under Application No.ZJUT-2023-117. All subjects were provided written informed consent. We used a Myo armband manufactured by Thalmic Labs to collect eight-channel sEMG and three-axis ACC signals. The sampling frequency of sEMG and ACC signal is 200Hz and 50Hz, respectively. Data transmission and processing are facilitated by connecting the Myo armband to the computer via Bluetooth. The experimental protocol proceeded as follows: Firstly, the subjects' skin was cleaned with alcohol to reduce impedance. Then, subjects executed prescribed gestures based on the movement instructions displayed on the screen. sEMG and ACC data were collected using the Myo armband positioned on the right forearm, as shown in Fig. 1(a). Each subject performed 11 distinct gestures, encompassing pointing index, victory, flexion of ring and little fingers, thumb clasp, palm open, fist, wrist flexion, wrist extension, wrist ulnar deviation, wrist radial deviation, and rest, as demonstrated in Fig. 1(b). Each gesture lasted for 6 seconds and was repeated 20 times, with a 5-second rest interval between consecutive gestures to mitigate muscle fatigue. Furthermore, the acquired sEMG and ACC data are presented in Fig. 1(c).

TABLE I
DETAILS OF THE TWO DATASETS

Name	Number of gestures	Number of subjects	Number of sEMG channels	Number of ACC channels	Number of Myo armband	Validation method
MyoData	11	17	8	3	1	Hold-out
Ninapro DB5	53	10	16	3	2	Subject-dependent

Ninapro-DB5: This dataset comprises sEMG and ACC data collected from 10 subjects. The subjects executed 53 distinct gestures, encompassing 12 finger movements, 17 wrist movements, 23 grasping and functional movements, and a rest state. Each gesture comprised six trials, repeated six times, with each trial lasting 5 seconds and a 3-second rest interval between consecutive trials. The experiment utilized two Myo armbands to simultaneously collect 16-channel sEMG and three-axis ACC signals, both sampled at 200Hz. For further details regarding Ninapro DB5, please consult the study [9].

More details are provided in Table I. In general, the two datasets differ in the number of gestures, subjects, sEMG channels, and ACC channels. Specifically, the MyoData dataset uses one Myo armband, while the Ninapro DB5 dataset uses two. Additionally, the two datasets employ different validation methods: MyoData uses hold-out validation, whereas Ninapro DB5 uses subject-dependent validation. Section III provides a detailed explanation of these two validation methods.

B. Preprocessing

During the signal preprocessing stage, the raw sEMG and ACC signals are filtered using a first-order Butterworth low-pass filter with a cutoff frequency of 1 Hz to remove high-frequency noise components and derive the signal envelope. Furthermore, to align with the data processing methodology of Ninapro-DB5 [9], the sampling frequencies of both sEMG and ACC signals in the MyoData dataset are adjusted to be consistent through resampling. To further mitigate the adverse effects of random noise in the signals, a moving average method is employed to smooth both signals:

$$y(n) = \frac{\sum_{i=0}^{m-1} x(n-i)}{m} \tag{1}$$

where $x(n)$ represents the sequence of the extracted signal envelope, $m = 10$ represents the degree of smoothing and $y(n)$ represents the sequence of the signal after smoothing. Given the real-time constraints of gesture recognition methods, it is typical to opt for an analysis window shorter than 300 ms [35]. In this study, a sliding window technique with a length of 200 ms and a step size of 25 ms is employed to extract sample data from the preprocessed sEMG and ACC signals, as depicted in Fig. 1(c).

III. METHODS

A. Problem Definition

This paper primarily focuses on the task of gesture recognition based on multi-modal signals from armband sensor, treating it as a multi-feature fusion classification problem. Multi-feature fusion classification involves integrating multi-domain feature information from different modalities to construct a highly accurate and robust gesture classification model. Specifically, $D = \{X_i, Y_i\}$ represents the armband-based gesture recognition dataset containing sEMG and ACC signals. The input $X_i = \left\{x_1^1, \ldots, x_n^m, \ldots, x_q^p | 1 \le m \le p, 1 \le n \le q\right\}$ represents the i-th sample, where x_n^m denotes the m-th domain feature of the n-th modality, and Y_i represents the corresponding class label. The goal of multi-feature fusion classification is to establish a feature encoding function $\mathcal{F}(\cdot)$, a feature fusion function $\Phi(\cdot)$ and a mapping function $\Gamma(\cdot)$ within D. Here, $\mathcal{F}(\cdot)$ enables sufficient encoding of the input X_i over multiple domains. $\Phi(\cdot)$ achieves the fusion of multi-modal and multi-domain information within the samples, retaining more discriminative fused features. Subsequently, the fused features are mapped to an output space to obtain the predicted label $\Gamma(\Phi(\mathcal{F}(X_i))) \in Y_i$ for the sample X_i. In this study, two modal signals (sEMG and ACC) and three feature domains (time-space-frequency) are utilized to construct a robust classification model capable of accurately recognizing gestures.

B. Model Architecture

This paper introduces a deep learning framework named MFN-SPSCA for effectively modelling the time-space-frequency patterns of sEMG and ACC signals to enhance gesture recognition accuracy, as depicted in Fig. 2. Initially, we design an RMSCM comprising a multi-scale convolution (MSC) submodule to capture short-term temporal pattern and employs a CNN-BiLSTM submodule to learn long-term temporal pattern. Subsequently, STGCN-SPS is employed to characterize the potential spatial features of multi-channel sEMG signals. Additionally, the CWT-CM is devised to extract frequency-domain features from both sEMG and ACC signals. Considering that frequency-domain and spatial features can complement time-domain features by offering deeper insights into the signal's oscillatory characteristics and the coordination of muscle groups. Therefore, we integrate the spatial and frequency-domain features using concatenation into auxiliary features. The auxiliary features, along with the long- and short-term temporal features, are fed into the CAFM, which enhances the intrinsic correlations among features, leading to more discriminative fused features. Finally, the fused features are passed into the CM to determine the predicted gesture category. Next, we will provide a comprehensive description of each module within the proposed MFN-SPSCA framework.

1) RMSCM: It consists of an MSC submodule and a CNN-BiLSTM submodule, as illustrated in Fig. 2(b). Specifically, assuming the input is represented as $x \in \mathbb{R}^{N\times 1\times T\times V}$,

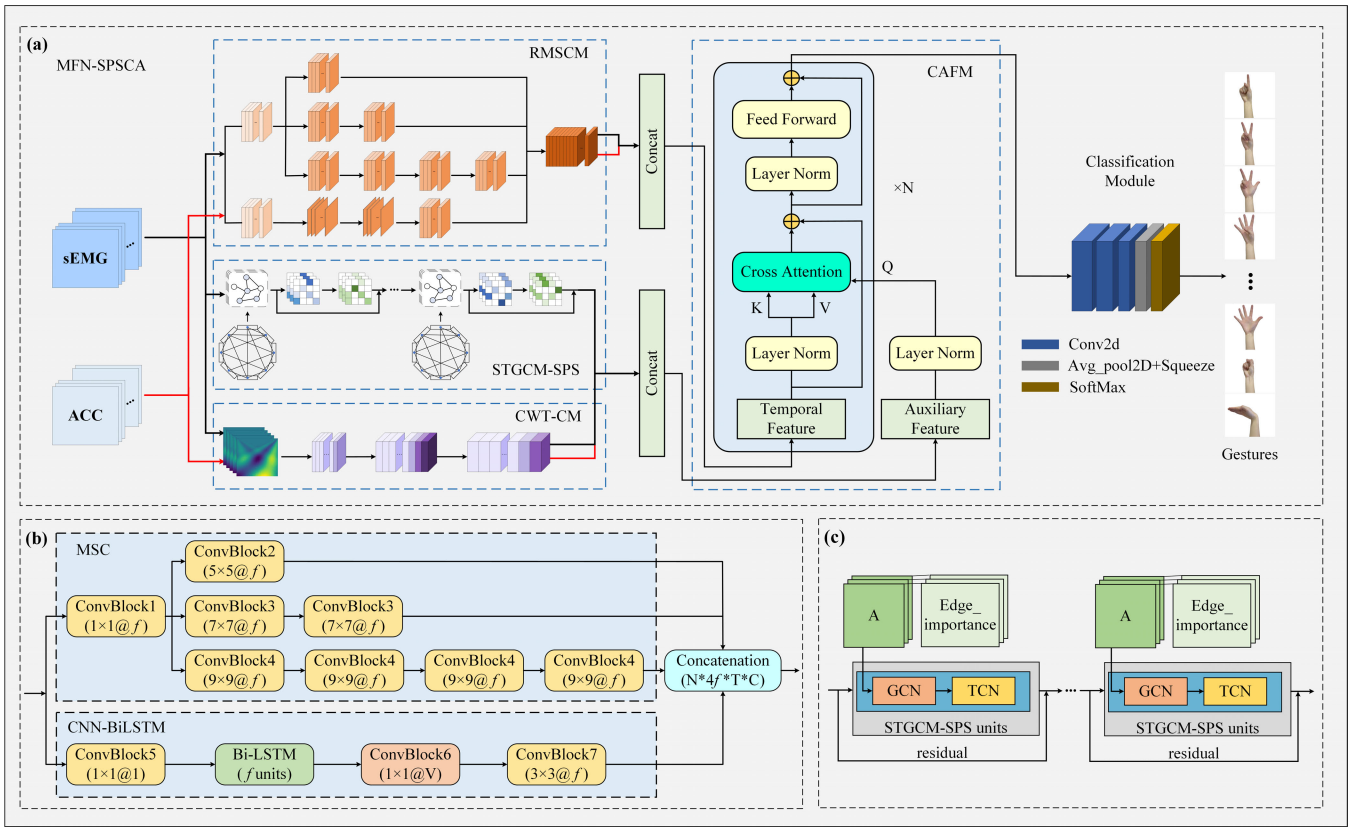


Fig. 2. Illustration of the proposed MFN-SPSCA framework. (a) MFN-SPSCA. (b) RMSCM. (c) STGCM-SPS.

where N represents the sample size, 1 represents the initial input dimension of RMSCM layer, T represents the time dimension, and V signifies the number of channels. Within the MSC submodule, ConvBlock1 is initially employed to extract shallow features ST from the preprocessed sEMG and ACC signals. Subsequently, ConvBlock2, ConvBlock3, and ConvBlock4 are utilized to conduct multi-scale convolution operations on the features ST, yielding short-term temporal features $ST_{b1} \in \mathbb{R}^{N\times f\times T\times V}$, $ST_{b2} \in \mathbb{R}^{N\times f\times T\times V}$, and $ST_{b3} \in \mathbb{R}^{N\times f\times T\times V}$, respectively. Furthermore, the CNN-BiLSTM submodule consists of ConvBlock5, a Bi-LSTM layer, ConvBlock6, and ConvBlock7. Initially, the ConvBlock5 is utilized to extract shallow features, followed by the integration of a core Bi-LSTM layer (with units denoted as f) to enhance the model's capability to capture long-term dependency. ConvBlock6 and ConvBlock7 are then employed to extract high-level long-term temporal features and ensure consistency in dimensions with short-term temporal features, thereby obtaining features $LT = \mathbb{R}^{N\times f\times T\times V}$. Each ConvBlock in the RMSCM is characterized by parameters "$k \times k@f$", where $k \times k$ represents the convolution kernel size, f ($f = 64$) denotes the convolution dimension, and relevant details are depicted in Fig. 2(b).

Next, the short-term temporal features (ST_{b1}, ST_{b2}, and ST_{b3}) and the long-term temporal features (LT) are fused via a concatenation operation to yield the long-short term temporal features $LST \in \mathbb{R}^{N\times 4f\times T\times V}$:

$$LST = \text{Concat}\left(ST_{b1}, ST_{b2}, ST_{b3}, LT\right) \tag{2}$$

where Concat $(\cdot)$ denotes the concatenation operation. In this study, the preprocessed sEMG and ACC signals are individually fed into the RMSCM, yielding features $LST_E \in \mathbb{R}^{N\times 4f\times T\times V_E}$ and $LST_A \in \mathbb{R}^{N\times 4f\times T\times V_A}$. The concatenation operation is employed to merge LST_E and LST_A, resulting in the final long- and short-term temporal features $LST^* \in \mathbb{R}^{N\times 4f\times T\times V'}$, $V' = V_E + V_A$:

$$LST^* = \text{Concat}\left(LST_E, LST_A\right) \tag{3}$$

2) STGCM-SPS: Recognizing gesture movements relies heavily on the functional connectivity among electrodes on the Myo armband. This study introduces a multi-layer STGCM-SPS based on a spatial partitioning strategy to capture this functional connectivity information. Initially, each sEMG channel is treated as a node in the graph, with connections between different channels represented as edges. Specifically, an undirected weighted graph $G = (V, A)$ is used in this paper, where $V = \{v_1, v_2, \ldots, v_n\}$ denotes the vertex set with n nodes, and $A = \left(a_{ij}\right)_{n\times n}$ is the adjacency matrix describing the edge weights between the nodes in V. Each element a_{ij} in A indicates the importance of the connection from node i to node j. Given fixed number and placement of sEMG electrodes, constructing an appropriate adjacency matrix is crucial. Here, we employ the Pearson correlation coefficient (PCC) to assess the functional connectivity between any two electrode channels, with the calculation equation as follows:

$$\rho_{ij} = \left|\frac{\text{Cov}\left(s_i, s_j\right)}{\sigma_i \sigma_j}\right| \tag{4}$$

where ρ_{ij} and $\mathrm{Cov}(i, j)$ represent the PCC and covariance between signal s_i and s_j, respectively. σ_i and σ_j represent the standard deviations of signal s_i and s_j, respectively. The value of ranges from 0 to 1, where $\rho_{ij} = 0$ signifies no correlation between nodes, and $\rho_{ij} = 1$ represents complete correlation. Additionally, a threshold τ is set to determine the elements a_{ij} in the adjacency matrix A:

$$a_{ij} = \begin{cases} 1, & if\ \rho_{ij} \geq \tau \\ 0, & if\ \rho_{ij} < \tau \end{cases} \tag{5}$$

Utilizing A to further calculate the Laplacian matrix L, as shown below:

$$L = \hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} \tag{6}$$

where $\hat{A} = A + I_N$ represents the normalized adjacency matrix, and $\hat{D} = \mathrm{diag}([d_1, d_2, \ldots, d_i, \ldots, d_N])$ is the degree matrix of A, $d_i = \sum_j a_{ij}$. The updating equation for each layer of GCN can be defined as follows:

$$Z_S = \sigma(LXW_S) \in \mathbb{R}^{N \times f_g \times T \times V} \tag{7}$$

where $X = [x_1, \ldots, x_N] \in \mathbb{R}^{N \times 1 \times T \times V}$ denotes the input, T represents the time dimension, V is the number of nodes, f_g represents the output dimension, $W_S \in \mathbb{R}^{V \times V}$ denotes the weight matrix of GCN, and $\sigma(\cdot)$ denotes the ReLU activation function. Additionally, we integrate a temporal convolutional block and residual connections into the GCN framework, resulting in the updating equation for each layer of the spatiotemporal graph convolution module (STGCM):

$$Z = \mathrm{Conv}(Z_S) + X, Z \in \mathbb{R}^{N \times f_{st} \times T \times V} \tag{8}$$

where f_{st} represents the output dimension.

3) Partitioning Strategy: According to Eq. (8), the parameters in graph convolution operations are shared, neglecting the influence of distinct armband sensor regions on edge weight parameters. To address this issue and enhance the high-level spatial feature representation capability of GCN, four spatial partitioning strategies are proposed: Uni-labeling, bi-labeling, tri-labeling, and multi-labeling partitioning strategies. Considering one of the nodes as the root node, depicted in Fig. 3(b), where red node denotes the root node v_i, blue nodes connected to v_i represent neighboring nodes v_j $(j = 1, 2 \cdots, n_s)$, and v_i and v_j form the neighborhood set $B(v_i)$. Subsequently, we introduce partitioning strategies to divide $B(v_i)$ into K subsets. Then, we can map the nodes in $B(v_i)$ to corresponding subset labels through the mapping $l_i : B(v_i) \rightarrow \{1, \ldots, K\}$. Therefore, the weight parameters can be expressed as follows:

$$w(v_i, v_j) = w'(l_i(v_j)) \tag{9}$$

a) Uni-labeling: In this strategy, the neighborhood set $B(v_i)$ is considered as a single entity. The execution of this strategy is equivalent to computing the inner product of the same weight vector with feature vectors on corresponding nodes. As shown in Fig. 3(a), both v_i and v_j are in red. Therefore, all nodes within the neighborhood set $B(v_i)$ are assigned the same weight. This strategy is characterized by its simplicity and low computational cost, making it suitable for handling relatively simple muscle activation patterns. In this case, $K = 1$, and $l_i(v_j) = 1, \forall i, j \in V$.

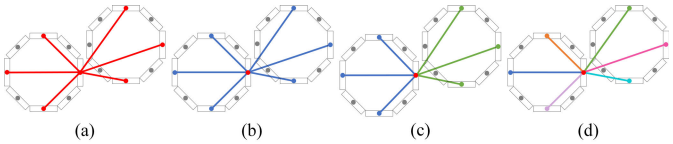


Fig. 3. Multiple partitioning strategies. (a) Uni-labeling partitioning strategy. (b) Bi-labeling partitioning strategy. (c) Tri-labeling partitioning strategy. (d) Multi-labeling partitioning strategy.

b) Bi-labeling: This strategy divides the neighborhood set $B(v_i)$ into two subsets: 1) the root node v_i itself, 2) the neighboring nodes v_j connected to the root node. Assuming the distance between the neighboring nodes v_j and root node v_i is $d_{i,j}$, where $d_{i,j} = 0$ represents the root node itself, and the remaining neighboring nodes are in the $d_{i,j} = 1$ subset. As shown in Fig. 3(b), the root node is red, and the neighboring nodes are blue. Therefore, the neighborhood set $B(v_i)$ is composed of two distinct weight vectors, achieving the modeling of local electrode interactions. Compared to the uni-labeling strategy, the bi-labeling partitioning strategy captures differential muscle activation patterns between the root node and its neighboring nodes. However, it struggles to capture complex interactions between muscle groups. Formally, we have $K = 2$ and $l_i(v_j) = d_{i,j}$.

c) Tri-labeling: Considering the spatial distribution characteristics of the Myo armband electrodes, we can divide the neighborhood set $B(v_i)$ into three subsets: 1) the root node v_i itself, 2) the neighboring nodes v_j connected to the root node v_i and the nodes v_i and v_j belong to the same Myo armband, 3) the neighboring nodes v_j connected to the root node v_i and the nodes v_i and v_j belong to the different Myo armband. This strategy, motivated by the significant differences in correlations between electrodes on different armbands, effectively represents complex muscle activation patterns and captures region-level activations involving multiple muscle groups. As shown in Fig. 3(c), the root node v_i is red, the neighboring nodes v_j are blue, and the neighboring nodes v_j are green. Formally, we have $K = 3$ and $l_i(v_j)$ is calculated as follows:

$$l_i(v_j) = \begin{cases} 1, & if\ v_j = v_i \\ 2, & if\ v_j \neq v_i,\ v_j \in B(v_i),\ and\ v_j \circledcirc v_i \\ 3, & if\ v_j \neq v_i,\ v_j \in B(v_i),\ and\ v_j \ominus v_i \end{cases} \tag{10}$$

where $\circledcirc$ represents that two nodes belong to the same armband, and $\ominus$ represents that two nodes belong to different armbands.

d) Multi-labeling: Considering the correlated interactions between any connected electrodes, we divide the neighborhood set $B(v_i)$ into N_s $(N_s = n_s + 1)$ subsets. As shown in Fig. 3(d), the root node is red, and each v_j has a different color. The advantage of this strategy lies in its ability to simultaneously capture the coordinated activity of multiple muscle groups. However, it is prone to information redundancy or excessive sensitivity to local muscle activations, and it is

less effective at capturing region-level muscle activity patterns. In this case, $K = N_s$ and $l_i\left(v_j\right)$ is calculated as follows:

$$l_i\left(v_j\right) = \begin{cases} 1, & if\ v_j = v_i \\ j, & if\ v_j \neq v_i,\ j \in n_s \end{cases} \tag{11}$$

The visual representation of the four partitioning strategies is illustrated in Fig. 3. In the Section III, we evaluate the effectiveness of the proposed partitioning strategies. More advanced partitioning strategies are expected to enhance modeling capabilities and recognition performance. By integrating graph structure methods with partitioning strategies, the update process of each layer in STGCM-SPS is transformed into:

$$S = \mathrm{Conv}\left(LXW_S'\left(l_i\left(v_j\right)\right)\right) + X,\ S \in \mathbb{R}^{N\times fst\times T\times V} \tag{12}$$

where f_{st} represents the output dimension. In this study, we utilize a 9-layer STGCM-SPS to extract spatial features from the preprocessed sEMG signal X. The transformation process is denoted as $S_m \in \left\{\mathbb{R}^{N\times f_{st}\times T\times V} | m = 1, 2, \ldots, 9\right\}$, where S_m denotes the output of the m-th layer of STGCM-SPS, and $f_{st} = 4f$.

4) CWT-CM: The preprocessed signals undergo continuous wavelet transform using the morlet wavelet basis function before being input into a convolution module to learn frequency-domain features. The CWT-CM structure comprises three convolutional layers, with a batch normalization (BN) layer and a ReLU activation function between each consecutive pair of convolutional layers. The convolutional layers have kernel sizes of 1×1, 3×3, and 3×3, with corresponding dimensions of 16, 64, and 128, respectively. Additionally, a convolution stride of 1 is set for the convolutional layers, and a padding scheme is applied to keep the size of the feature map unchanged. As a result, after applying the CWT-CM, features denoted as $F^* \in \mathbb{R}^{N\times 2f\times T\times V}$ are obtained, with the calculation equation as follows:

$$F^* = \mathrm{CM}\left(\mathrm{CWT}\left(X\right)\right) \in \mathbb{R}^{N\times 2f\times T\times V} \tag{13}$$

where $X = [x_1, \ldots, x_N] \in \mathbb{R}^{N\times 1\times T\times V}$ represents the input, $\mathrm{CM}(\cdot)$ denotes the convolution module operation, and $\mathrm{CWT}(\cdot)$ stands for the continuous wavelet transform function. Therefore, the preprocessed sEMG and ACC signals are separately fed into the CWT-CM module, yielding frequency-domain features $F_E^* \in \mathbb{R}^{N\times 2f\times T\times V}$ for sEMG and $F_A^* \in \mathbb{R}^{N\times 2f\times T\times V}$ for ACC. The features F_A^* and F_E^* are subsequently concatenated with to derive the fused auxiliary feature $AF^* \in \mathbb{R}^{N\times 4f\times T\times V}$:

$$AF^* = \mathrm{Concat}\left(F_E^*, F_A^*, S_m\right) \tag{14}$$

5) CAFM: Through the above steps, we get the temporal features LST^* and auxiliary features AF^*. As previously mentioned, conventional feature-level fusion methods struggle to effectively capture finer-grained correlation information among multiple features. Here, we introduce the CAFM based on cross-attention fusion method, which the attention weight of each modality in cross-attention fusion is guided by other modality, and the complementarity among different modality features are exploited to extract more representative features, thereby achieving effective fusion of multi-modal and time-space-frequency information. According to Eq. (3) and Eq. (14), the obtained long- and short-term temporal features and auxiliary features are reshaped into $X_{LST^*} \in \mathbb{R}^{N'\times T\times V}$ and $X_{AF^*} \in \mathbb{R}^{N'\times T\times V}$. Next, the query (Q_s), key (K_t) and value (V_t) are generated by linear projection on X_{LST^*} and X_{AF^*}, respectively. Finally, a cross-attention matrix (CA) is generated. The description of the CAFM is outlined below:

$$\left(Q_s, K_t, V_t\right) = \left(X_{LST^*}W^q, X_{AF^*}W^k, X_{AF^*}W^v\right) \tag{15}$$

$$\begin{aligned} CA^* &= \mathrm{Attention}\left(Q_s, K_t, V_t\right) \\ &= \mathrm{Soft\,max}\left(\frac{Q_s \times K_t^T}{\sqrt{d}}\right)V_t \end{aligned} \tag{16}$$

where $W^q \in \mathbb{R}^{v\times d}$, $W^k \in \mathbb{R}^{v\times d}$, $W^v \in \mathbb{R}^{v\times d}$ are the weight matrices of linear projection, v represents the feature dimension, and d denotes the dimension of tokens in embedding space. The cross-attention matrix measures the attention scores of temporal features from the perspective of auxiliary features. We set the number of layers of the CAFM to $N = 2$, the final result of the cross-attention fusion feature is obtained:

$$Res_{CA^*} = \mathrm{LN}\left(CA^*\right) + LST^* \tag{17}$$

$$F_{CA^*} = Res_{CA^*} + \mathrm{FFN}\left(\mathrm{LN}\left(Res_{CA^*}\right)\right) \tag{18}$$

where $\mathrm{FFN}(\cdot)$ is a feedforward network implemented by a linear layer with an output dimension of Res_{CA^*}, and $\mathrm{LN}(\cdot)$ represents the layer normalization.

6) CM: Finally, we employ three residual convolutional layers and pooling layers as the CM (as shown in Fig. 2). By inputting F_{CA^*} into CM, we derive the final high-level feature F_{CM}. The convolutional layers have filter sizes of 3×3 and strides of 2, while those within the residual structure have filter sizes of 1×1 and strides of 2. Subsequently, the output is fed into the Softmax layer to acquire the predicted labels $\hat{z} \in \mathbb{R}^N$:

$$\begin{cases} F_{CM} = \mathrm{Conv}^*\left(\mathrm{Avgpool}\left(F_{CA^*}\right)\right) \\ \hat{z} = \mathrm{Soft\,max}\left(F_{CM}\right) \end{cases} \tag{19}$$

where $\mathrm{Conv}^*(\cdot)$ represents three convolution operations, $\mathrm{Avgpool}(\cdot)$ represents average pooling operation, and $\hat{z} \in \mathbb{R}^N$ represents the predicted labels.

C. Model Parameter Settings

Suppose a labeled sEMG-ACC-based gesture recognition dataset $\{X, Y, Z\} = \{x_i, y_i, z_i\}_{i=1}^N$ is given, where $X \in \mathbb{R}^{N\times 1\times T\times V}$ represents multi-channel sEMG signals, $Y \in \mathbb{R}^{N\times 1\times T\times V}$ represents three-axis acceleration signals, and $Z \in \mathbb{R}^N$ denotes the true labels. Firstly, the long- and short-term temporal features LST^* are obtained by processing x_i and y_i using RMSCM according to Eq. (2)-(3). Subsequently, x_i and y_i are transformed to the fused auxiliary features AF^* using STGCM-SPS and CWT-CM based on Eq. (4)-(14). Afterwards, the features F_{CA^*} are obtained by using CAFM to fuse LST^* and AF^* based on Eq. (15)-(18). Finally, F_{CA^*} is input into the CM according to Eq. (19) to obtain the ultimate predicted labels $\hat{z}_i \in \mathbb{R}^N$. Algorithm 1 provides detailed steps for the optimization process of the model.

TABLE II
PERFORMANCE METRICS OF THE MFN-SPSCA MODEL FOR DIFFERENT DATASETS

Dataset	Accuracy	Precision	Recall	F1-score	Kappa	Inference time(ms)	Parameter sizes(MB)
MyoData	98.78%	98.88%	98.81%	0.9880	0.9865	12.18	37.22
Ninapro DB5	91.60%	91.72%	91.62%	0.9161	0.9138	21.25	37.56

In this paper, cross-entropy loss L is used to assess the inconsistency between the true labels z_i and the predicted labels $\hat{z}_i$. The calculation equation is as follows:

$$L = -\frac{1}{M}\sum_{k=1}^{K}\sum_{i=1}^{M} z_i^k \log\left(\hat{z}_i^k\right) + \alpha\|\Theta\|_1 + \beta\|A\|_1 \tag{20}$$

where α and β are constants, and Θ represents all hyper-parameters during the model training process. The training process utilizes the SGD optimizer for its entirety, encompassing 40 epochs, initialized with a learning rate of 0.01, and employing a batch size of 64. All networks in this paper are implemented using Python 3.9 and PyTorch 2.0.1. Both training and evaluation are performed on an NVIDIA GeForce GPU (RTX 4060).

Algorithm 1 The Optimizing Procedure of the MFN-SPSCA

Input: A labeled gesture recognition dataset based on sEMG and ACC signals $\{X, Y, Z\} = \{x_i, y_i, z_i\}_{i=1}^{N}$, the number of epoch E_P, the batch size of each epoch B_a, and model hyper-parameters Θ.
Output: The optimal set of the model parameter Θ.
Initialize: The parameters Θ in the proposed MFN-SPSCA model.
for $e = 1 : E_p$ **do**
  for $b = 1 : B_a$ **do**
    Sample one batch size of samples $x_{e,b}$, $y_{e,b}$ and $z_{e,b}$ from $\{X, Y, Z\}$.
    Calculate the long- and short-term temporal features LST^* by feeding $x_{e,b}$ and $y_{e,b}$ into the RMSCM according to Eq. (2)-(3).
    Calculate the fused auxiliary features AF^* by feeding $x_{e,b}$ and $y_{e,b}$ into the STGCM-SPS and CWT-CM according to Eq. (4)-(14).
    Calculate the F_{CA^*} by feeding LST^* and AF^* into the CAFM based on Eq. (15)-(18).
    Calculate the prediction label $\hat{z}_i$ by feeding F_{CA^*} into the CM based on Eq. (19).
    Utilize $\hat{z}_i$ to calculate the loss function according to Eq. (20).
    Update the model parameters Θ via SGD optimizer based on the loss function.
  end for
end for

D. Experimental Setup

We follow the data partitioning strategy outlined in studies [31], [37] for the Ninapro DB5 dataset, and a hold-out validation method is employed. This entails the 1st, 3rd, 4th, and 6th repetitions of each gesture as the training set, and the 2nd and 5th repetitions serve as the test set. As for the MyoData dataset, a subject-dependent validation method is used, where all samples of any subject are divided into the training and test sets in a 7:3 ratio. Furthermore, we utilize five metrics to evaluate the performance of MFN-SPSCA model: Accuracy, Precision, Recall, Kappa, and F1-score:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{21}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{22}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{23}$$

$$\text{Kappa} = \frac{P_o - P_e}{1 - P_e} \tag{24}$$

$$\text{F1} - \text{score} = \frac{2\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{25}$$

where TP, TN, FP, and FN correspond to true positive, true negative, false positive, and false negative, respectively. P_o denotes predictive consistency, and P_e represents chance consistency.

IV. RESULTS

A. Overall Performance

As depicted in Table II, the MFN-SPSCA model demonstrates excellent recognition performance on both the MyoData and Ninapro DB5 dataset, achieving recognition accuracies of 98.78% and 91.60% on MyoData and Ninapro DB5, respectively. Specifically, on the MyoData dataset, the MFN-SPSCA model attains precision, recall, and F1-score values of 0.9888, 0.9881, and 0.9880, respectively. Similarly, on Ninapro DB5 dataset, the MFN-SPSCA model achieves precision, recall, and F1-score values of 0.9172, 0.9162, and 0.9161, respectively. To comprehensively illustrate the classification ability of MFN-SPSCA for each gesture, we compute the confusion matrices for both the MyoData and Ninapro DB5 datasets, as shown in Fig. 4. It is evident from the results that MFN-SPSCA achieves classification accuracy of exceeding 95% for all gestures in MyoData, with no significant confusion observed. In the Ninapro DB5 dataset, apart from the recognition accuracy of the 2nd and 10th gestures being below 80%, the classification accuracy of most gestures is above 90%.

Furthermore, the decoding time is a crucial metric for assessing the practical applicability of a model. Each test sample from the MyoData and Ninapro DB5 dataset was individually inputted into the MFN-SPSCA model to calculate the average inference time. Table II reveals that the average inference times per sample for MyoData and Ninapro DB5 datasets are 12.18 ms and 21.25 ms, respectively. Therefore,

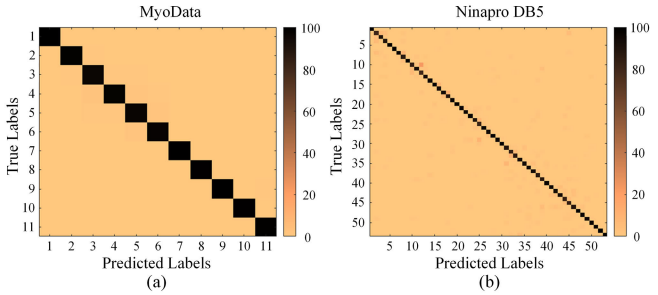


Fig. 4. Confusion matrices of the MFN-SPSCA model on different datasets. (a) MyoData dataset. (b) Ninapro DB5 dataset.

the inference time of the MFN-SPSCA model falls below the time constraints for real-time myoelectric control (300 ms) [36], suggesting its potential application in real-time HGR systems. In our experiments, when the model was run on standard hardware configurations, the model parameter sizes were 37.22 MB for the MyoData dataset and 37.56 MB for the Ninapro DB5 dataset.

B. Analysis of Threshold Parameters and Spatial Partitioning Strategies

The size of the threshold parameter τ and the choice of spatial partitioning strategy significantly influence the recognition performance of the MFN-SPSCA model. Hence, we thoroughly examine the impact of these hyperparameters in this section. Specifically, we vary the threshold parameter τ across values of 0.2, 0.4, 0.6, and 0.8. For spatial partitioning strategies, we apply the uni-labeling and bi-labeling partitioning strategies to the MyoData dataset (utilizing a single armband), and the uni-labeling, bi-labeling, tri-labeling, and multi-labeling partitioning strategies to the Ninapro DB5 dataset.

Fig. 5. illustrates the impact of different threshold parameters and spatial partitioning strategies on the classification performance of the model. For the MyoData dataset, under any threshold parameter τ, the bi-labeling partitioning strategy consistently outperforms the uni-labeling partitioning strategy. At threshold parameter $\tau = 0.2$, the lowest loss value and highest accuracy value are observed, while at threshold parameter $\tau = 0.8$, the highest loss and lowest accuracy value are observed, indicating a gradual decrease in model performance with increasing threshold. Similarly, for the Ninapro DB5 dataset, under any partitioning strategy, a smaller threshold parameter τ correlates with improved model performance. Additionally, the performance of the tri-labeling partitioning strategy surpasses that of the uni-labeling, bi-labeling, and multi-labeling partitioning strategies. The uni-labeling partitioning strategy shows the weakest performance, primarily because all nodes share the same weights, limiting its ability to capture spatial relationships between different nodes. In contrast, the bi-labeling partitioning strategy assigns different weights to neighboring nodes and the root node, allowing it to capture the differential spatial relationships between these two types of nodes, resulting in better performance than the uni-label strategy. The tri-labeling partitioning strategy

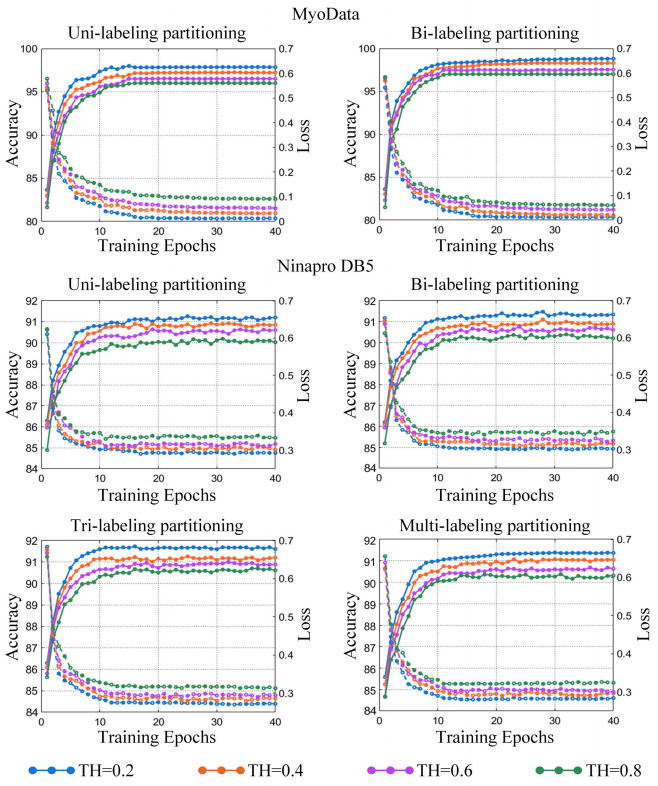


Fig. 5. Comparison of the model's accuracy and loss under different threshold parameters and spatial partitioning strategies.

accounts for more complex muscle activation patterns, particularly by representing interactions both within and between the armband nodes. Meanwhile, the tri-labeling partitioning strategy achieves the best performance, indicating that muscle activation patterns vary across armbands and that it provides better modeling of region-level muscle activation patterns. Although the multi-labeling partitioning strategy captures the coordinated activities of multiple muscle groups, it is prone to information redundancy and, due to its inability to capture region-level spatial semantic features, performs worse than the tri-labeling partitioning strategy.

C. Ablation Experiment

In this subsection, we conduct ablation experiments to evaluate the validity and significance of proposed functional modules in the MFN-SPSCA model. The compared models include: (1) MFN-SPSCA w/o RMSCM: RMSCM removed. (2) MFN-SPSCA w/o STGCM-SPS: STGCM-SPS module removed. (3) MFN-SPSCA w/o CWT-CM: CWT-CM removed. (4) MFN-SPSCA w/o CFAM: CFAM removed and replaced with concatenation operation. (5) MFN-SPSCA: Our proposed model.

The results of the ablation experiments indicate that, compared to MFN-SPSCA w/o RMSCM, MFN-SPSCA w/o STGCM-SPS, MFN-SPSCA w/o CWT-CM, and MFN-SPSCA w/o CAFM, the MFN-SPSCA model achieves the highest accuracy on both MyoData and Ninapro DB5 datasets, as depicted in Fig. 6. Therefore, the RMSCM, STGCM-SPS, CWT-CM, and CAFM in MFN-SPSCA all make significant

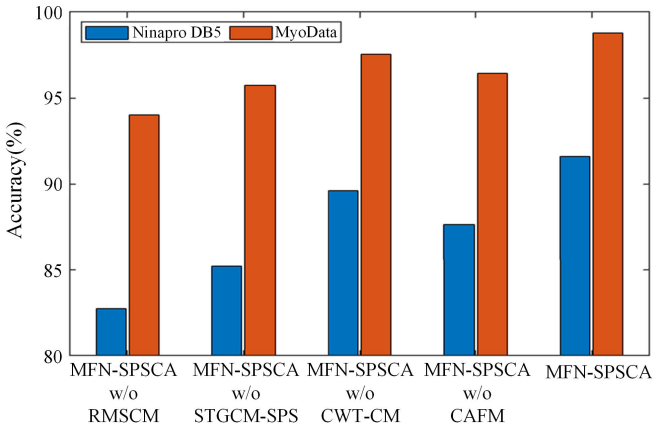


Fig. 6. Comparison of accuracy for ablation studies on MyoData and Ninapro DB5 dataset.

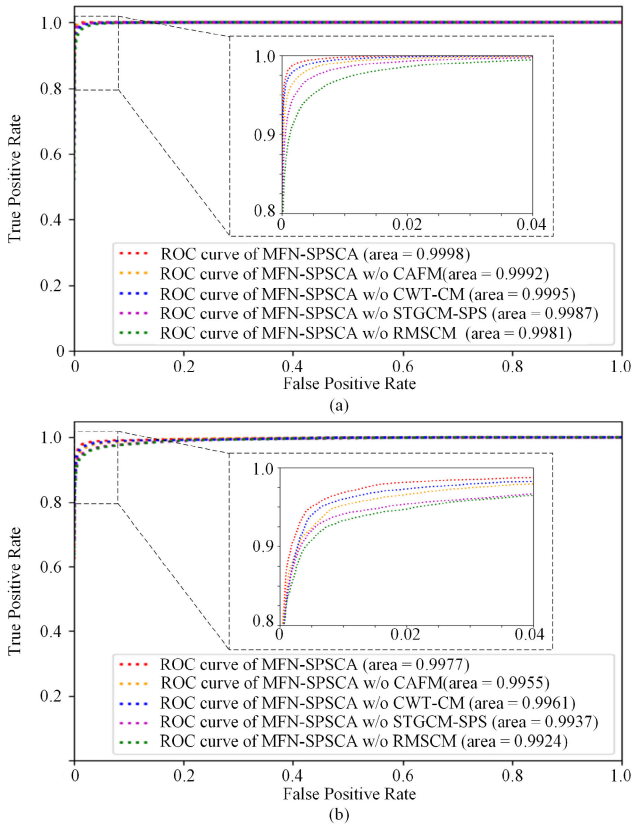


Fig. 7. Comparison of AUC curves for ablation studies on different datasets. (a) MyoData dataset. (b) Ninapro DB5 dataset.

contributions. Omitting any functional module would impact the classification performance of the MFN-SPSCA model. Furthermore, the RMSCM has the most substantial impact on overall performance improvement, resulting in an accuracy increase of 1.89% and 3.05% on the MyoData and Ninapro DB5 datasets, respectively.

Additionally, we analyzed the receiver operating characteristic (ROC) curves of the five ablation models, as shown in Fig. 7(a) and 7(b). The MFN-SPSCA model exhibits the highest area under the curve (AUC) value of 0.9998 on

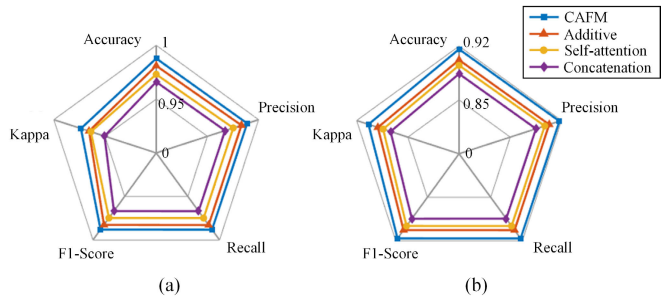


Fig. 8. Comparison of fusion strategies studies on different datasets. (a) MyoData dataset. (b) Ninapro DB5 dataset.

the MyoData dataset, surpassing MFN-SPSCA w/o RMSCM, MFN-SPSCA w/o STGCM-SPS, MFN-SPSCA w/o CWT-CM, and MFN-SPSCA w/o CAFM by 0.0017, 0.0011, 0.0003, and 0.0006, respectively. Similarly, on the Ninapro DB5 dataset, the MFN-SPSCA model also shows the highest AUC value of 0.9977, exceeding MFN-SPSCA w/o RMSCM, MFN-SPSCA w/o STGCM-SPS, MFN-SPSCA w/o CWT-CM, and MFN-SPSCA w/o CAFM by 0.0053, 0.0040, 0.0016, and 0.0022, respectively. In summary, the MFN-SPSCA model demonstrates the best performance of gesture recognition, with all four functional modules (RMSCM, STGCM-SPS, CWT-CM, and CAFM) contributing significantly to its improvement.

V. Discussion

A. Fusion Method Comparison

In this study, we assess the effectiveness of our proposed fusion strategy by comparing it with three mainstream feature-level fusion methods: Concatenation fusion [18], additive fusion [32], and self-attention fusion [42]. As illustrated in Fig. 8, The CAFM outperforms all others in terms of Accuracy, Precision, Recall, F1-score, and Kappa. On the MyoData dataset, CAFM achieves higher accuracy compared to concatenation fusion, additive fusion, and self-attention fusion methods by 2.18%, 0.68%, and 1.46%, respectively. Similarly, on the Ninapro DB5 dataset, CAFM achieves higher accuracy compared to concatenation fusion, additive fusion, and self-attention fusion methods by 1.42%, 0.53%, and 0.87%, respectively. Hence, CAFM demonstrates stronger effectiveness than mainstream fusion methods in integrating multi-modal and time-space-frequency domain feature information. This superiority can be attributed to CAFM's layer-by-layer learning method, which fully exploits the intricate dependencies and interactions among feature domains.

B. t-SNE

This paper employs t-SNE technique to visualize the spatial distribution of high-level features extracted by MFN-SPSCA in the classification tasks of MyoData and Ninapro DB5, aiming to intuitively depict the discriminative capability of these features. The larger the distance between clusters of different feature types, and the stronger the cohesion within clusters of the same feature type, the better the feature distinctiveness [38]. Fig. 9 depicts the spatial distribution of feature clusters for the 1st, 3rd, 7th, and 10th gestures in the MyoData

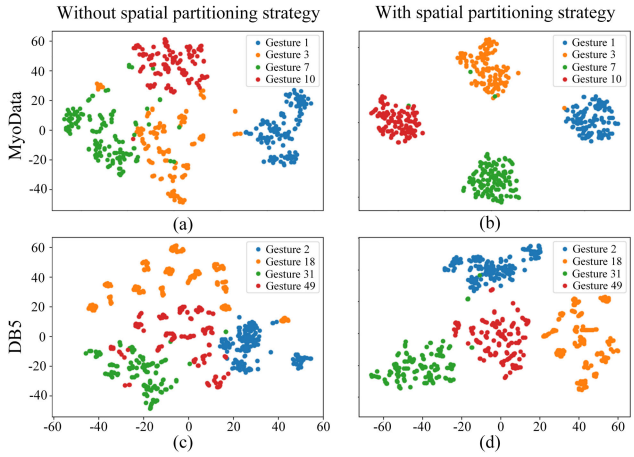


Fig. 9. t-SNE maps. Different colors represent different categories.

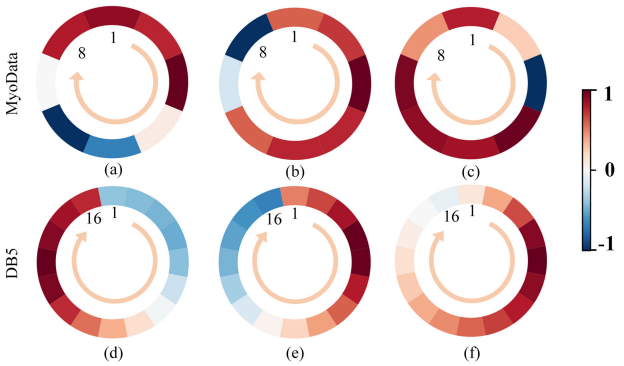


Fig. 10. CAM maps for different gesture actions in the two datasets.

dataset and the 2nd, 18th, 31st, and 49th gestures in the Ninapro DB5 dataset with and without the spatial partitioning strategy, respectively. As shown in Fig. 9(a) and 9(c), without spatial partitioning strategies, significant overlap between different feature clusters is evident. However, upon introducing spatial partitioning strategies, it is notable that the distances between different feature clusters increase, and the clustering within the same feature cluster improves, as illustrated in Fig. 9(b) and 9(d).

C. Spatial Features of CAM

To visually demonstrate the spatial representations learned by MFN-SPSCA, we use class activation mapping (CAM) for visualization, as shown in Fig. 10. The CAM technique is applied to detect the activation patterns of the sEMG signal channels that the MFN-SPSCA model focuses on, represented in circular diagrams where the channels are arranged clockwise, with "1" and "8" indicating the first and eighth channels, respectively. In particular, red indicate stronger activation levels, and blue indicates weaker activation levels. Figs. 10(a), 10(b), and 10(c) represent the average activation patterns for all channels corresponding to the 3rd, 7th, and 9th gestures in the MyoData dataset, while Figs. 10(d), 10(e), and 10(f) show the patterns for the 3rd, 15th, and 47th gestures in the Ninapro DB5 dataset. It is evident that the MFN-SPSCA model learns distinct spatial representations for different types of gestures across both datasets. Also, this visualization provides further insights into muscle activation patterns and neuromuscular control mechanisms during the execution of gestures.

D. Comparison With State-of-the-Art Methods

To assess the performance superiority of the MFN-SPSCA model, we conducted comprehensive comparisons on the MyoData dataset with six mainstream models: MV-CNN [31], Attention model [39], MLP [40], ICA-CNN [41], HyFusion [42], and CLCNet [37]. Similarly, on the Ninapro DB5 dataset, we compared the MFN-SPSCA model with six mainstream models: MV-CNN [31], CNN+LSTM [20], Attention model [39], EELM [43], SE-CNN [44], and CLCNet [37].

TABLE III
COMPARISON RESULTS WITH MAINSTREAM MODELS

Method	Year	Dataset	Window size	Number of geastures	Acc
MV-CNN [31]	2019	MyoData	200ms	11	92.58%
Attention model [40]	2020		260ms	11	87.43%
MLP [41]	2021		200ms	11	82.54%
ICA-CNN [42]	2023		200ms	11	86.74%
HyFusion [43]	2023		200ms	11	93.78%
CLCNet [38]	2023		160ms	11	92.18%
Ours	-		160ms	11	**98.65%**
			200ms	11	**98.78%**
MV-CNN [31]	2019	DB5	200ms	41	91.31%
CNN+LSTM [20](w/o IMU)	2020		260ms	53	85.60%
Attention model [40]	2020		260ms	53	91.00%
EELM [44](w/o IMU)	2022		300ms	52	77.90%
SE-CNN [45](w/o IMU)	2023		260ms	53	87.42%
CLCNet [38]	2023		160ms	18	92.16%
Ours	-		160ms	18	**94.48%**
			200ms	41	**91.75%**
			200ms	52	**91.50%**
			200ms	53	**91.60%**

Due to variations in window lengths and the number of gesture categories across studies, comparative experiments were conducted on the MyoData and Ninapro DB5 datasets with window lengths of 160 ms and 200 ms, respectively. Additionally, for the Ninapro DB5 dataset, comparative experiments were performed with gesture counts of 18, 41, 52, and 53 to ensure comprehensive coverage. Naturally, the comparative experiments are based on the following consensus: The smaller the window length and the greater the number of gesture classes, the greater the challenge for the model's recognition accuracy.

The comparative results are shown in Table III. For the MyoData dataset, the MFN-SPSCA model achieves higher recognition accuracy compared to the contrastive models. Specifically, with a sliding window length of 160 ms, the MFN-SPSCA model outperforms the approach in [37] on the MyoData dataset by 6.47%. For the Ninapro DB5 dataset, the MFN-SPSCA model also exhibits higher recognition accuracy compared to the contrastive models. In the classification task of 53 gesture categories using smaller sliding window lengths, the MFN-SPSCA model achieves recognition accuracy improvements of 6%, 0.6%, and 4.18% over the

approaches in [20], [39], and [44], respectively. Additionally, in the classification task with 52 gesture classes, the MFN-SPSCA model with a smaller sliding window length parameter achieves a recognition accuracy improvement of 13.60% compared to the approach in [43]. In the classification tasks with 41 and 18 gesture categories under the same sliding window length parameter, the MFN-SPSCA model achieves recognition accuracy improvements of 0.44% and 2.32% over the approaches in [31] and [37], respectively.

VI. Conclusion and Future Works

This paper proposes a multi-feature fusion network named MFN-SPSCA based on sEMG and ACC signals. The MFN-SPSCA model shows significant promise in enhancing the performance of patient-operated assistive devices, such as exoskeletons and prosthetics, by providing more accurate and robust gesture decoding. Specially, the MFN-SPSCA model comprises RMSCM, STGCM-SPS, CWT-CM, and CAFM. Firstly, the MFN-SPSCA model extracts long- and short-term temporal features from sEMG and ACC signals using RMSCM. Secondly, it employs STGCM-SPS to construct a graph structure among the armband to explore potential spatial dependencies of different electrodes. Thirdly, it utilizes time-frequency maps and CWT-CM to extract frequency domain features from sEMG and ACC. Finally, CAFM is employed to fuse time-space-frequency features, and CM is used for gesture category prediction. This study employs a method that combines mathematical statistics and performance evaluation to comprehensively evaluate the classification accuracy and robustness of the MFN-SPSCA model from the aspects of overall performance, hyperparameter comparison, and ablation research based on the sEMG and ACC signals collected by armband. Extensive performance comparison experiments are conducted on two datasets: MyoData and Ninapro DB5. The MFN-SPSCA model achieves the highest recognition accuracy compared to state-of-the-art methods, reaching 98.78% and 91.60% on MyoData and Ninapro DB5, respectively. These results demonstrate that the proposed MFN-SPSCA model exhibits excellent recognition performance and generalization capability. Additionally, the proposed spatial partitioning strategy effectively focuses on spatial feature representation among different regions, enhancing the ability of STGCM-SPS to mine spatial information. Simultaneously, the proposed fusion strategy concentrates more precisely on potential dependencies among features, achieving effective fusion of multi-modal and multi-feature information. It is hoped that this work can accurately recognize common gestures and be applied to human-computer interaction, motion rehabilitation, and virtual reality.

Nevertheless, the proposed MFN-SPSCA model does not fully account for physiological differences between individuals, limiting its generalization to unseen subjects. Variations in individual physiology can alter sEMG signal patterns, thereby affecting model performance. At present, the MFN-SPSCA model relies on data from known individuals and lacks the ability to accommodate cross-subject variability. In future work, we aim to explore domain adaptation and domain generalization algorithms to improve the model's adaptability and generalization across different individuals. This will enhance the model's robustness in real-world applications, allowing it to maintain high performance across a diverse user population and advance the development of myoelectric prosthesis technology.

References

[1] L. E. Sucar, F. Orihuela-Espina, R. L. Velazquez, D. J. Reinkensmeyer, R. Leder, and J. Hernández-Franco, "Gesture therapy: An upper limb virtual reality-based motor rehabilitation platform," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 3, pp. 634–643, May 2014.

[2] D. Farina et al., "The extraction of neural information from the surface EMG for the control of upper-limb prostheses: Emerging avenues and challenges," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 4, pp. 797–809, Jul. 2014.

[3] J. Zhang, J. Dai, S. Chen, G. Xu, and X. Gao, "Design of finger exoskeleton rehabilitation robot using the flexible joint and the MYO armband," in *Proc. Intell. Robot. Appl.*, vol. 11745, 2019, pp. 213–225.

[4] Z. Ren, J. Yuan, J. Meng, and Z. Zhang, "Robust part-based hand gesture recognition using Kinect sensor," *IEEE Trans. Multimedia*, vol. 15, no. 5, pp. 1110–1120, Aug. 2013.

[5] T. Zhao, J. Liu, Y. Wang, H. Liu, and Y. Chen, "Towards low-cost sign language gesture recognition leveraging wearables," *IEEE Trans. Mobile Comput.*, vol. 20, no. 4, pp. 1685–1701, Apr. 2021.

[6] K. Li, J. Zhang, L. Wang, M. Zhang, J. Li, and S. Bao, "A review of the key technologies for sEMG-based human–robot interaction systems," *Biomed. Signal Process. Control*, vol. 62, Sep. 2020, Art. no. 102074.

[7] T. Bao, Y. Zhao, S. A. R. Zaidi, S. Xie, P. Yang, and Z. Zhang, "A deep Kalman filter network for hand kinematics estimation using sEMG," *Pattern Recognit. Lett.*, vol. 143, pp. 88–94, Mar. 2021.

[8] G. Biagetti, P. Crippa, S. Orcioni, and C. Turchetti, "Homomorphic deconvolution for MUAP estimation from surface EMG signals," *IEEE J. Biomed. Health Informat.*, vol. 21, no. 2, pp. 328–338, Mar. 2017.

[9] S. Pizzolato, L. Tagliapietra, M. Cognolato, M. Reggiani, H. Müller, and M. Atzori, "Comparison of six electromyography acquisition setups on hand movement classification tasks," *PLoS ONE*, vol. 12, no. 10, Oct. 2017, Art. no. e0186132.

[10] B. Zhu, D. Zhang, Y. Chu, Y. Gu, and X. Zhao, "SeNic: An open source dataset for sEMG-based gesture recognition in non-ideal conditions," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 1252–1260, 2022.

[11] A. Fougner, E. Scheme, A. D. C. Chan, K. Englehart, and Ø. Stavdahl, "A multi-modal approach for hand motion classification using surface EMG and accelerometers," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Aug. 2011, pp. 4247–4250.

[12] A. Gijsberts and B. Caputo, "Exploiting accelerometers to improve movement classification for prosthetics," in *Proc. IEEE 13th Int. Conf. Rehabil. Robot. (ICORR)*, Jun. 2013, pp. 1–5.

[13] M. Atzori, A. Gijsberts, H. Müller, and B. Caputo, "Classification of hand movements in amputated subjects by sEMG and accelerometers," in *Proc. 36th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Chicago, IL, USA, Aug. 2014, pp. 3545–3549.

[14] E. Gokgoz and A. Subasi, "Comparison of decision tree algorithms for EMG signal classification using DWT," *Biomed. Signal Process. Control*, vol. 18, pp. 138–144, Apr. 2015.

[15] A. Krasoulis, I. Kyranou, M. S. Erden, K. Nazarpour, and S. Vijayakumar, "Improved prosthetic hand control with concurrent use of myoelectric and inertial measurements," *J. NeuroEng. Rehabil.*, vol. 14, no. 1, pp. 1–14, Dec. 2017.

[16] I. Kuzborskij, A. Gijsberts, and B. Caputo, "On the challenge of classifying 52 hand movements from surface electromyography," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Aug. 2012, pp. 4931–4937.

[17] A. Gijsberts, M. Atzori, C. Castellini, H. Müller, and B. Caputo, "Movement error rate for evaluation of machine learning methods for sEMG-based hand movement classification," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 4, pp. 735–744, Jul. 2014.

[18] W. Wei, Y. Wong, Y. Du, Y. Hu, M. Kankanhalli, and W. Geng, "A multi-stream convolutional neural network for sEMG-based gesture recognition in muscle-computer interface," *Pattern Recognit. Lett.*, vol. 119, pp. 131–138, Mar. 2019.

[19] C. Wang, W. Guo, H. Zhang, L. Guo, C. Huang, and C. Lin, "sEMG-based continuous estimation of grasp movements by long-short term memory network," *Biomed. Signal Process. Control*, vol. 59, May 2020, Art. no. 101774.

[20] R. N. Khushaba, A. Phinyomark, A. H. Al-Timemy, and E. Scheme, "Recursive multi-signal temporal fusions with attention mechanism improves EMG feature extraction," *IEEE Trans. Artif. Intell.*, vol. 1, no. 2, pp. 139–150, Oct. 2020.

[21] M. Atzori, M. Cognolato, and H. Müller, "Deep learning with convolutional neural networks applied to electromyography data: A resource for the classification of movements for prosthetic hands," *Frontiers Neurorobot.*, vol. 10, p. 9, Sep. 2016.

[22] S. Shen, K. Gu, X.-R. Chen, C.-X. Lv, and R.-C. Wang, "Gesture recognition through sEMG with wearable device based on deep learning," *Mobile Netw. Appl.*, vol. 25, no. 6, pp. 2447–2458, Dec. 2020.

[23] P. Koch, M. Dreier, M. Böhme, M. Maass, H. Phan, and A. Mertins, "Inhomogeneously stacked RNN for recognizing hand gestures from magnetometer data," in *Proc. 27th Eur. Signal Process. Conf. (EUSIPCO)*, A Coruna, Spain, Sep. 2019, pp. 1–5.

[24] W. Wang, B. Chen, P. Xia, J. Hu, and Y. Peng, "Sensor fusion for myoelectric control based on deep learning with recurrent convolutional neural networks," *Artif. Organs*, vol. 42, no. 9, pp. E272–E282, Sep. 2018.

[25] X. Luo et al., "Synergistic myoelectrical activities of forearm muscles improving robust recognition of multi-fingered gestures," *Sensors*, vol. 19, no. 3, p. 610, Feb. 2019.

[26] H. Wang, L. Xu, A. Bezerianos, C. Chen, and Z. Zhang, "Linking attention-based multiscale CNN with dynamical GCN for driving fatigue detection," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021.

[27] F. Hu, L. Zhang, X. Yang, and W.-A. Zhang, "EEG-based driver fatigue detection using spatio-temporal fusion network with brain region partitioning strategy," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 8, pp. 9618–9630, Aug. 2024.

[28] L. Shen, M. Sun, Q. Li, B. Li, Z. Pan, and J. Lei, "Multiscale temporal self-attention and dynamical graph convolution hybrid network for EEG-based stereogram recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 1191–1202, 2022.

[29] P. K. Atrey, M. A. Hossain, A. E. Saddik, and M. S. Kankanhalli, "Multimodal fusion for multimedia analysis: A survey," *Multimedia Syst.*, vol. 16, no. 6, pp. 345–379, Nov. 2010.

[30] Y. Li, X. Chen, X. Zhang, K. Wang, and Z. J. Wang, "A sign-component-based framework for Chinese sign language recognition using accelerometer and sEMG data," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 10, pp. 2695–2704, Oct. 2012.

[31] W. Wei, Q. Dai, Y. Wong, Y. Hu, M. Kankanhalli, and W. Geng, "Surface-electromyography-based gesture recognition by multi-view deep learning," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 10, pp. 2964–2973, Oct. 2019.

[32] U. Côté-Allard et al., "Deep learning for electromyographic hand gesture signal classification using transfer learning," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 4, pp. 760–771, Apr. 2019.

[33] J. Liu, W. Chen, M. Li, and X. Kang, "Continuous recognition of multifunctional finger and wrist movements in amputee subjects based on sEMG and accelerometry," *Open Biomed. Eng. J.*, vol. 10, no. 1, pp. 101–110, Nov. 2016.

[34] S. Duan, L. Wu, A. Liu, and X. Chen, "Alignment-enhanced interactive fusion model for complete and incomplete multimodal hand gesture recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 4661–4671, 2023.

[35] H. Mizuno, N. Tsujiuchi, and T. Koizumi, "Forearm motion discrimination technique using real-time EMG signals," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Aug. 2011, pp. 4435–4438.

[36] A. Krasoulis, S. Vijayakumar, and K. Nazarpour, "Multi-grip classification-based prosthesis control with two EMG-IMU sensors," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 2, pp. 508–518, Feb. 2020.

[37] L. Wang, J. Fu, H. Chen, and B. Zheng, "Hand gesture recognition using smooth wavelet packet transformation and hybrid CNN based on surface EMG and accelerometer signal," *Biomed. Signal Process. Control*, vol. 86, Sep. 2023, Art. no. 105141.

[38] A. Narayan, B. Berger, and H. Cho, "Assessing single-cell transcriptomic variability through density-preserving data visualization," *Nature Biotechnol.*, vol. 39, no. 6, pp. 765–774, Jun. 2021.

[39] D. Josephs, C. Drake, A. Heroy, and J. Santerre, "sEMG gesture recognition with a simple model of attention," in *Proc. Mach. Learn. Health*, 2020, pp. 126–138.

[40] J. O. de Oliveira de Souza, M. D. Bloedow, F. C. Rubo, R. M. de Figueiredo, G. Pessin, and S. J. Rigo, "Investigation of different approaches to real-time control of prosthetic hands with electromyography signals," *IEEE Sensors J.*, vol. 21, no. 18, pp. 20674–20684, Sep. 2021.

[41] S. Shen, X. Wang, M. Wu, K. Gu, X. Chen, and X. Geng, "ICA-CNN: Gesture recognition using CNN with improved channel attention mechanism and multimodal signals," *IEEE Sensors J.*, vol. 23, no. 4, pp. 4052–4059, Feb. 2023.

[42] S. Duan, L. Wu, B. Xue, A. Liu, R. Qian, and X. Chen, "A hybrid multimodal fusion framework for sEMG-ACC-based hand gesture recognition," *IEEE Sensors J.*, vol. 23, no. 3, pp. 2773–2782, Feb. 2023.

[43] F. Peng et al., "Gesture recognition by ensemble extreme learning machine based on surface electromyography signals," *Frontiers Human Neurosci.*, vol. 16, Jun. 2022, Art. no. 911204.

[44] Z. Xu et al., "A novel SE-CNN attention architecture for sEMG-based hand gesture recognition," *Comput. Model. Eng. Sci.*, vol. 134, no. 1, pp. 157–177, 2023.